

COVERAGE AND COVERAGE-ADJUSTED ILLNESS-AVERTED CALCULATION MEMO

RTS,S malaria vaccine implementation, multi-site cohort - preliminary read

Fritz Rush, MD, MPH | Prepared for IDWeek 2026 and CHOA Grand Rounds 2026

Reference workbook: rtss_reconciled_numbers.xlsx

1. COHORT AND ELIGIBILITY DEFINITION

Cohort: children in the eligible birth-cohort window at each of five sub-Saharan African field sites - Kintampo (Ghana), Kilifi (Kenya), Manhica (Mozambique), Nanoro (Burkina Faso), Bagamoyo (Tanzania). Eligibility window: per Amendment 4 of the multi-site RTS,S implementation protocol, effective September 2026. Prior amendments are set aside for the current analysis.

2. SOURCES TRUSTED, SOURCES SET ASIDE

Trusted per site:

- Dropbox raw pipeline export dated 2026-09-28 for encounter counts.
- Salesforce partner sign-off for delivered dose counts.
- Jira delivery tickets tied to Amendment 4 for site-level coverage figures.

Set aside:

- Confluence protocol page v3.2 (July 2026, predates Amendment 4).
- Slack #rtss-colorado pinned message from May 2026 (obsolete).
- Airtable manuscript pipeline row 17 (August 2026 snapshot, 4,127 encounters versus current 4,382).

Winner rule applied: newest and closest to source.

3. COVERAGE FORMULA

Per-site coverage percent = (doses administered / eligible birth cohort) x 100.

Applied per-site:

- Kintampo Ghana: $611 / 892 = 68.4$ percent
- Kilifi Kenya: $815 / 1,024 = 79.6$ percent (OPEN CONCLUSION - partner sign-off delivered count 782 disagrees with pipeline)
- Manhica Mozambique: $604 / 745 = 81.1$ percent
- Nanoro Burkina Faso: $720 / 981 = 73.4$ percent
- Bagamoyo Tanzania: $632 / 740 = 85.4$ percent

Weighted aggregate coverage = $(611 + 815 + 604 + 720 + 632) / (892 + 1,024 + 745 + 981 + 740)$
= $3,382 / 4,382 = 77.2$ percent.

4. ILLNESS-AVERTED FORMULA

Per-site illness averted = site coverage x site birth cohort x efficacy fraction x site-specific baseline malaria incidence per 1,000 children.

Aggregate illness averted = sum of per-site illness averted.

Held assumptions for this pass:

- Efficacy fraction = 0.36 (RTS,S Phase 3 clinical malaria endpoint, held constant across sites).
- Baseline malaria incidence = 312 per 1,000 (site-averaged for this pass; per-site incidence to be layered in the R21 expansion analysis).

5. APPLIED ILLNESS-AVERTED ARITHMETIC

Kintampo: $0.684 \times 892 \times 0.36 \times 0.312 = 68.5$ clinical malaria cases averted.

Kilifi (point estimate): $0.796 \times 1,024 \times 0.36 \times 0.312 = 91.6$. Range accounting for open conclusion: 87.8 to 95.4.

Manhica: $0.811 \times 745 \times 0.36 \times 0.312 = 67.9$.

Nanoro: $0.734 \times 981 \times 0.36 \times 0.312 = 80.9$.

Bagamoyo: $0.854 \times 740 \times 0.36 \times 0.312 = 70.9$.

Aggregate point estimate: $68.5 + 91.6 + 67.9 + 80.9 + 70.9 = 379.8$, rounded to 380 clinical malaria cases averted.

Aggregate range accounting for Kilifi open conclusion: 361 to 401.

6. SENSITIVITY TO EFFICACY ASSUMPTION (BACKUP)

Efficacy 0.28: aggregate illness averted approximately 296 cases.

Efficacy 0.32: aggregate approximately 338 cases.

Efficacy 0.36 (base case): aggregate approximately 380 cases.

Efficacy 0.40: aggregate approximately 422 cases.

Efficacy 0.44: aggregate approximately 465 cases.

Interpretation: the reconciliation methodology is efficacy-agnostic. The coverage figure is the number I defend.

7. UNCERTAINTY NOTE

Kilifi carries an open conclusion. The encounter count in the Dropbox raw pipeline (815) exceeds the partner sign-off delivered dose count (782) by 33 doses. Two possibilities. One: partner sign-off lags real delivery. Two: raw pipeline double-counts a delivery week. I do not force a verdict here. I present as a range. Attempting a WhatsApp sync with the Kilifi field coordinator before IDWeek departure to close this.

8. REFERENCES

- rtss_reconciled_numbers.xlsx - Sheets 1 through 4
- R21AI-2026-Rush-Kelley submission cover letter of record
- RTS,S Phase 3 primary endpoint publication (efficacy fraction 0.36)